Rich (Chen-Feng) Tsai

rich.tsai1103@gmail.com | (510) 679-7136 | chenfengtsai.github.io

RESEARCH INTERESTS

Generative models for music, audio and sequential data; autoregressive and multi-codebook architectures; multi-task learning and knowledge distillation; robustness and long-horizon generation.

EDUCATION

University of Chicago Dec 2023
M.S. in Applied Data Science — GPA 3.97 / 4.0

University of California, Berkeley May 2022
B.A. in Data Science; B.A. in Economics

PUBLICATIONS

- "Mitigating Structural Collapse in Multi-Codebook Autoregressive Music Generation via Structure-Aware Scheduled Sampling." | *International Society for Music Information Retrieval Conference (ISMIR)*, 2026.
- "Rethinking Correlated Tasks in Multi-Task World Model Distillation: Beyond Gradient Conflict." | *Under review at AISTATS*, 2026.
- "An Interactive Dashboard on Top-k Recommendation for Cross-model Evaluation." | *ACM SIGIR*, 2022.
- "Can Large Language Models Play Text Games Well? Current State-of-the-Art and Open Questions." | *arXiv preprint*, 2023.

RESEARCH EXPERIENCE

Research & Development Specialist Sep 2024 – Present
Institute of Information Science, Academia Sinica | Advisor: Mi-Yen Yeh

- Showed that structural collapse in multi-codebook autoregressive music models arises from how exposure bias compounds across the 2D RVQ grid during long-horizon generation, rather than from exposure bias alone.
- Proposed Structure-Aware Scheduled Sampling (SASS) for long-horizon music generation; compared with teacher-forced MusicGen, it reduced KL divergence over PaSST label distributions by 5.6–10.0% and improved beat regularity by up to 14.3%, with no loss in text-audio alignment (ISMIR 2026).
- Showed that negative transfer in multi-task world model distillation can stem from dynamics averaging, which gradient-conflict diagnostics miss; across 28 Meta-World task pairs, its cost tracked confusability-weighted dispersion (r = 0.90) (under review, AISTATS 2027).
- Designed COMET, a single teacher with orthogonalized experts and per-source dynamics heads, matching or exceeding the mean return of separate single-task teachers on four out-of-distribution targets with 3.5× fewer parameters.

Research Assistant Mar 2024 – Mar 2025
Helen Wills Neuroscience Institute, UC Berkeley | Advisor: Cécile Tissot

- Built a mixed-effects linear regression model to convert between tau protein PET tracers, improving cross-study comparability in Alzheimer's disease research.
- Analyzed protein transformation pathways across disease stages, supporting a 7% improvement in early-detection models.

Research Assistant Feb 2023 – Jun 2023
Toyota Technological Institute at Chicago | Advisor: Hongyuan Mei

- Engineered game-playing agents using GPT-3 and LangChain to autonomously explore and map text-based adventure environments (Zork), reducing manual mapping effort by 90%.

Research Assistant Mar 2021 – Mar 2022
Research Center for Information Technology Innovation (CITI), Academia Sinica | Advisor: Chuan-Ju Wang

- Deployed a Bayesian Personalized Ranking recommendation system for a video-streaming partner, increasing click-through rate by 23%; built a companion visualization API for comparative model evaluation.

MUSIC EXPERIENCE

Music Composer & Sound Designer | Ultraviolet Studio Sep 2024 – Present

- Composed, mixed, and mastered 15+ dynamic background music tracks for the rhythm game Deemo using Ableton Live.
- Produced and edited original sound effects for the 78-episode Taiwanese anime DNA×CAT.

Musical Training and Performance

- **Associate Concertmaster,** Dongshan High School Orchestra; orchestra received a **Superior rating (特優), High School Division,** National Student Music Competition, Taiwan, 2016 and 2017.
- **Classical training:** Piano (since age 5) and violin (since age 8).
- **Additional practice:** Synthesizer (pop), jazz guitar, and electronic music production.

TECHNICAL PROJECTS

Drone Liquid Level Surveillance System Capstone Lead

C++, YOLOv5, GCP Vertex AI — Honorable Mention, UChicago Autumn 2023 Capstone Showcase

- Built an automated liquid-level monitoring system on a Tello drone with a YOLOv5 model fine-tuned on GCP to detect fill levels in production bottles, improving inventory-management efficiency.

TECHNICAL SKILLS

ML / DL: PyTorch, TensorFlow, scikit-learn
Languages: Python, C++, SQL, R, Java, Scala
Infrastructure & Tools: Git, Docker, GCP, AWS, Azure, Spark